QP CODE: MOD12102026

RegNo:.....

Name:.....

MAR AUGUSTHINOSE COLLEGE, RAMAPURAM

MGU-BCA(H) MODEL EXAMINATION, October 2026

MG3CCRBCA203–DESIGN AND ANALYSIS OF ALGORITHMS

Third Semester

Maximum Marks : 50 **Time : 1.5 Hours**

(Remember(K),Understand(U),Apply(A),Analyse(An))

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

PartA

Answer All questions. Each question carries 2 marks.

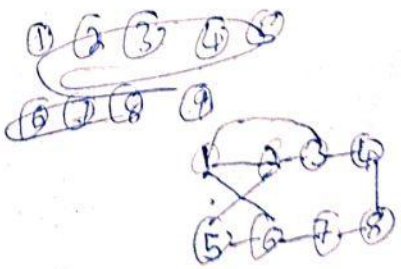
1.	Define control abstraction?	[k][1]
2.	Define implicit and explicit constraints.	[k][3]
3.	Differentiate between greedy and dynamic programming?.	[u][1]
4.	Explain the control abstraction of backtracking.	[u][1]
5.	Define NQueen's problem.	[k][1]
6.	Define optimal solution and feasible solution.	[k][2]
7.	Define Principle of Optimality.	[k][3]
8.	Define Spanning tree	[An][2,3]
9.	Write the objective function and feasibility constraints of knapsack problem.	[u][2]
10.	Define Live node, E-node & Dead node.	[k][3]

(10x2=20)

PartB

Answer any THREE questions.Each question carries 5 marks.

11.	Write the algorithm for all-pairs shortest path.	[u][3]
12.	ExplainTime complexity and Space Complexity	[u][3]
13.	Writean algorithm for Knapsack Problem	[u][3]
14.	Write an algorithm for binary search.	[u][2]
15.	Explain algorithm for DFS and BFS.	[u][3]
16.	Explain Hamiltonian cycle.	[u][3]



(3x5=15)

PartC

AnsweranyONEquestions.Eachquestioncarries15marks.

17.	Explain Sum of Subset Problem with example.	[u][2]
18.	Explain Prim's algorithm for finding MST.?	[u][2]

(1x15=15)